

Task – 1

```
UW PICO 5.09 File: Pratical-3.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    printf("Parent PID : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());
    fflush(stdout);

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");
        fflush(stdout);

        printf("Child is now waiting for 10 seconds...\n");
        fflush(stdout);

        sleep(10);

        printf("Child resumed running after waiting.\n");
        fflush(stdout);

        printf("Child terminating...\n");
        return 0;
    }
    else
    {
        // Parent process
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Parent PPID : %d\n", getppid());

        printf("Parent is running...\n");
        fflush(stdout);
    }
}
```

```
printf("Parent is waiting for the child to terminate...\n");
fflush(stdout);

wait(NULL);

printf("Child has terminated.\n");
printf("Parent terminating...\n");
}

return 0;
}
```

Compilation, Input & Output :

Task-2

Inputs and Outputs :

```
Parent process started.  
Parent PID : 1194  
Parent PPID : 1073  
  
--- PARENT PROCESS ---  
Parent PID : 1194  
Parent PPID : 1073  
Parent is running...  
Parent is waiting for the child to terminate...  
  
--- CHILD PROCESS ---  
Child PID : 1198  
Child PPID : 1194  
Child is running...  
Child is now waiting for 10 seconds...  
Child resumed running after waiting.  
Child terminating...  
Child has terminated.  
Parent terminating...  
boddanapurnachandarrao@boddanas-MacBook-Air Practical-3 %
```

```
718      305  0 09:53 pts/0      00:00:00 ./practical-3  
719      718  0 09:53 pts/0      00:00:00 ./practical-3  
721      702  0 09:53 pts/2      00:00:00 grep --color=auto practical-3
```

```
Child process terminating...  
Child terminated.  
Parent process terminating...
```

```
Name: practical-3
Umask: 0022
State: S (sleeping)
Tgid: 780
Ngid: 0
Pid: 780
PPid: 779
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1000 1000 1000 1000
FDSize: 64
Groups: 4 20 24 25 27 29 30 44 46 100 107 1000
NSTgid: 780
NSpid: 780
NSpgid: 779
NSsid: 305
Kthread: 0
VmPeak: 2692 kB
VmSize: 2692 kB
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 808 kB
VmRSS: 808 kB
RssAnon: 96 kB
RssFile: 712 kB
RssShmem: 0 kB
VmData: 224 kB
VmStk: 136 kB
VmExe: 4 kB
VmLib: 1748 kB
VmPTE: 48 kB
VmSwap: 0 kB
HugetlbPages: 0 kB
CoreDumping: 0
THP_enabled: 1
untag_mask: 0xffffffffffffffff
Threads: 1
SigQ: 1/31087
SigPnd: 0000000000000000
ShdPnd: 0000000000000000
SigBlk: 0000000000000000
SigIgn: 0000000000000000
```